Lab 3: Architecture Selection & Justification

Pharmacy Expiry & Re-order Dispatch Engine

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Architecture Selection: We chose **Layered Architecture** for the Pharmacy Expiry & Re-order Dispatch Engine.

Reason 1 – Clear separation of concerns matching system requirements

The system's requirements naturally split into three concerns: a pharmacy interface used by the Pharmacy Clerk for stock lookup, dispensing and inventory operations; core business rules such as batch-expiry tracking, FEFO selection, safety-threshold checking and purchase-order generation; and persistent storage of medicines, batches, stock levels, expiry dates and order records. A layered architecture maps directly onto this split — Presentation, Business and Data layers — making the design easy to understand, test and maintain without the added complexity of coordinating independently deployed services.

Reason 2 – Fits the scale and deployment model of a hospital pharmacy

The engine is intended for a hospital pharmacy where inventory operations, dispensing decisions and supplier ordering are part of one cohesive workflow. A Microservices style would introduce additional operational overhead such as service discovery, network communication and distributed data consistency. A simple Client-Server style is also too coarse-grained to clearly separate FEFO decision logic, stock threshold validation and supplier-order processing. Layered architecture provides the appropriate modularity while keeping the deployment straightforward.

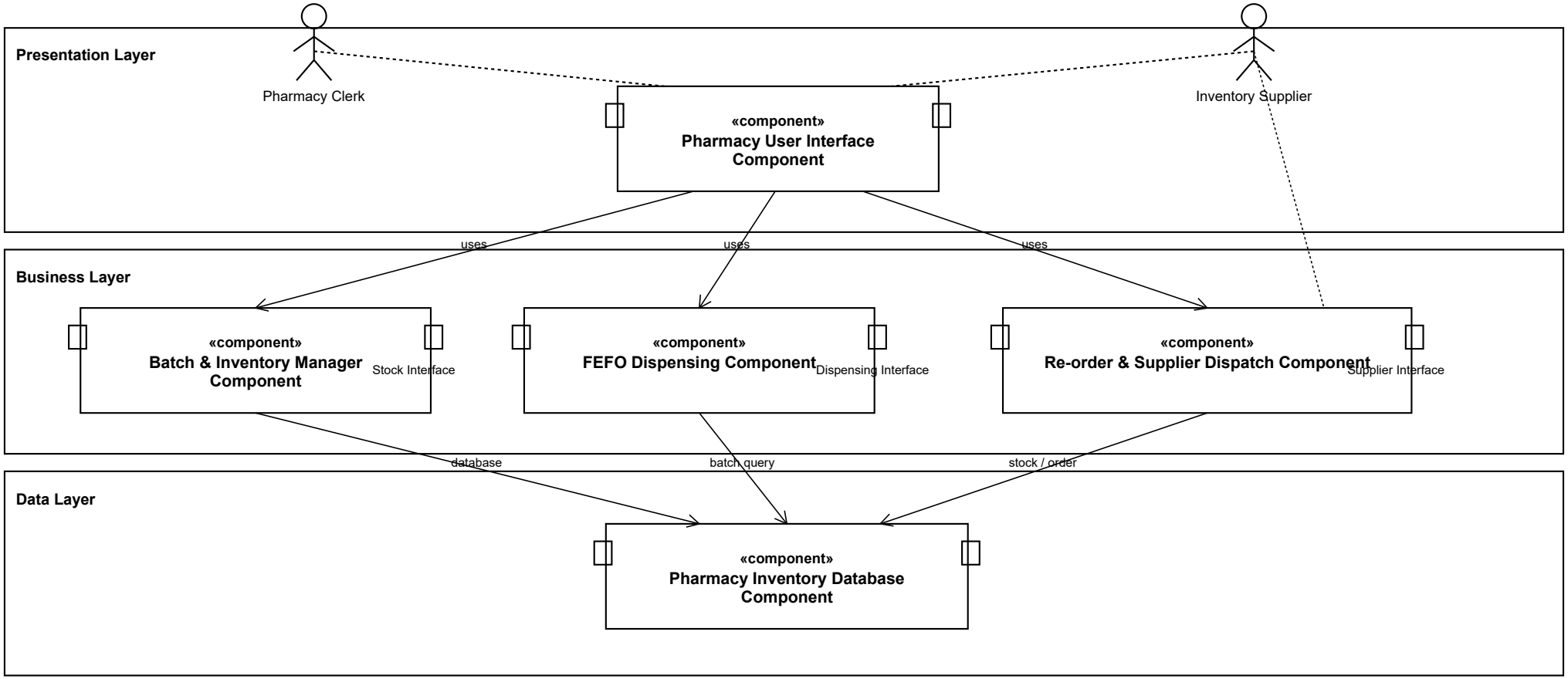
Security Advantage

The Presentation Layer does not communicate directly with the Data Layer. Every request passes through the Business Layer, where authentication, authorization and validation can be enforced. For example, the system can restrict sensitive inventory and supplier-order operations to authorized pharmacy staff. This controlled path reduces the attack surface and prevents malformed or unauthorized requests from directly changing medicine or order records.

Performance Benefit

The system must quickly identify the correct medicine batch during dispensing and react when stock falls below its safety threshold. With direct in-process communication between the Business and Data layers, the Batch & Inventory Manager can use indexed medicine/batch records to retrieve expiry and stock data efficiently. This avoids the additional network round trips and distributed coordination overhead that a microservices design would introduce for a single cohesive pharmacy workflow.

Pharmacy Expiry & Re-order Dispatch Engine — Layered Architecture



Architecture flow: The Pharmacy Clerk interacts with the Presentation Layer. Requests are routed to the Business Layer, where inventory/expiry rules, FEFO selection and re-order dispatch logic are applied. The Business Layer reads and updates medicine batches, expiry dates, stock levels and purchase-order records through the Data Layer. The Inventory Supplier receives purchase-order requests from the Re-order & Supplier Dispatch component.